

```
In [87]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

چی هستند؟ warnings اصلاً !)

پیامی که پایتون نمایش می‌دهد برای اینکه = (هشدار) warning

مشکلی «کوچک» در کد وجود دارد

یا در نسخه‌های آینده رفتار این خط عوض می‌شود

یا بهتر است کد را اصلاح کنید

هشدار = خطا نیست کد متوقف نمی‌شود ⚠

```
In [88]: # import warnings
# warnings.filterwarnings('ignore')
```

```
In [89]: df=pd.read_csv('Flipkart Mobile 2.csv')
```

```
In [90]: df
```

Out[90]:

	brand	model	base_color	processor	screen_size	ROM	RAM	display_size	num_re
--	-------	-------	------------	-----------	-------------	-----	-----	--------------	--------

0	Apple	iPhone SE	Black	Water	Very Small	64	2	4.7	
1	Apple	iPhone 12 Mini	Red	Ceramic	Small	64	4	5.4	
2	Apple	iPhone SE	Red	Water	Very Small	64	2	4.7	
3	Apple	iPhone XR	Others	iOS	Medium	64	3	6.1	
4	Apple	iPhone 12	Red	Ceramic	Medium	128	4	6.1	
...	...	...	...	...	...	...	...	...	
425	Xiaomi	Redmi 6 Pro	Black	Qualcomm	Small	32	3	5.8	
426	Xiaomi	Redmi 6 Pro	Red	Qualcomm	Small	64	4	5.8	
427	Xiaomi	Mi 11 Lite	Others	Qualcomm	Large	128	6	6.5	
428	Xiaomi	Redmi 8A Dual	Blue	Qualcomm	Medium	32	3	6.2	
429	Xiaomi	Redmi 6 Pro	Blue	Qualcomm	Small	32	3	5.8	

430 rows × 16 columns



In [91]: df.head()

Out[91]:

	brand	model	base_color	processor	screen_size	ROM	RAM	display_size	num_rear_c
0	Apple	iPhone SE	Black	Water	Very Small	64	2	4.7	
1	Apple	iPhone 12 Mini	Red	Ceramic	Small	64	4	5.4	
2	Apple	iPhone SE	Red	Water	Very Small	64	2	4.7	
3	Apple	iPhone XR	Others	iOS	Medium	64	3	6.1	
4	Apple	iPhone 12	Red	Ceramic	Medium	128	4	6.1	

In [92]: `df.shape`

Out[92]: (430, 16)

In [93]: `df.head(5)`

Out[93]:

	brand	model	base_color	processor	screen_size	ROM	RAM	display_size	num_rear_c
0	Apple	iPhone SE	Black	Water	Very Small	64	2	4.7	
1	Apple	iPhone 12 Mini	Red	Ceramic	Small	64	4	5.4	
2	Apple	iPhone SE	Red	Water	Very Small	64	2	4.7	
3	Apple	iPhone XR	Others	iOS	Medium	64	3	6.1	
4	Apple	iPhone 12	Red	Ceramic	Medium	128	4	6.1	

In [94]: `df.tail(5)`

Out[94]:

	brand	model	base_color	processor	screen_size	ROM	RAM	display_size	num_rear
--	-------	-------	------------	-----------	-------------	-----	-----	--------------	----------

425	Xiaomi	Redmi 6 Pro	Black	Qualcomm	Small	32	3	5.8	
426	Xiaomi	Redmi 6 Pro	Red	Qualcomm	Small	64	4	5.8	
427	Xiaomi	Mi 11 Lite	Others	Qualcomm	Large	128	6	6.5	
428	Xiaomi	Redmi 8A Dual	Blue	Qualcomm	Medium	32	3	6.2	
429	Xiaomi	Redmi 6 Pro	Blue	Qualcomm	Small	32	3	5.8	



In [95]: `df.shape`

Out[95]: (430, 16)

In [96]: `df.size`

Out[96]: 6880

In [97]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 430 entries, 0 to 429
Data columns (total 16 columns):
#   Column                Non-Null Count  Dtype
---  -
0   brand                  430 non-null   object
1   model                  430 non-null   object
2   base_color             430 non-null   object
3   processor              430 non-null   object
4   screen_size            430 non-null   object
5   ROM                    430 non-null   int64
6   RAM                    430 non-null   int64
7   display_size           430 non-null   float64
8   num_rear_camera        430 non-null   int64
9   num_front_camera       430 non-null   int64
10  battery_capacity        430 non-null   int64
11  ratings                 430 non-null   float64
12  num_of_ratings          430 non-null   int64
13  sales_price             430 non-null   int64
14  discount_percent        430 non-null   float64
15  sales                   430 non-null   float64
dtypes: float64(4), int64(7), object(5)
memory usage: 53.9+ KB
```

In [98]: `df.describe()`

Out[98]:

	ROM	RAM	display_size	num_rear_camera	num_front_camera	battery_c
count	430.000000	430.000000	430.000000	430.000000	430.000000	430
mean	105.748837	5.320930	6.369767	2.904651	1.044186	4529
std	63.164064	2.182635	0.369549	0.952350	0.227280	986
min	8.000000	1.000000	4.700000	1.000000	1.000000	1800
25%	64.000000	4.000000	6.300000	2.000000	1.000000	4000
50%	128.000000	4.000000	6.500000	3.000000	1.000000	4500
75%	128.000000	6.000000	6.500000	4.000000	1.000000	5000
max	512.000000	12.000000	7.600000	4.000000	3.000000	7000

In [99]: `df.describe()`

Out[99]:

	ROM	RAM	display_size	num_rear_camera	num_front_camera	battery_c
count	430.000000	430.000000	430.000000	430.000000	430.000000	430
mean	105.748837	5.320930	6.369767	2.904651	1.044186	4529
std	63.164064	2.182635	0.369549	0.952350	0.227280	986
min	8.000000	1.000000	4.700000	1.000000	1.000000	1800
25%	64.000000	4.000000	6.300000	2.000000	1.000000	4000
50%	128.000000	4.000000	6.500000	3.000000	1.000000	4500
75%	128.000000	6.000000	6.500000	4.000000	1.000000	5000
max	512.000000	12.000000	7.600000	4.000000	3.000000	7000

In [100... `df.columns`

Out[100... Index(['brand', 'model', 'base_color', 'processor', 'screen_size', 'ROM',
 'RAM', 'display_size', 'num_rear_camera', 'num_front_camera',
 'battery_capacity', 'ratings', 'num_of_ratings', 'sales_price',
 'discount_percent', 'sales'],
 dtype='object')

In [101... `df.dtypes`

```
Out[101... brand      object
model      object
base_color  object
processor   object
screen_size object
ROM         int64
RAM         int64
display_size float64
num_rear_camera int64
num_front_camera int64
battery_capacity int64
ratings     float64
num_of_ratings int64
sales_price  int64
discount_percent float64
sales        float64
dtype: object
```

```
In [102... df.isnull().sum()
```

```
Out[102... brand      0
model      0
base_color  0
processor   0
screen_size 0
ROM         0
RAM         0
display_size 0
num_rear_camera 0
num_front_camera 0
battery_capacity 0
ratings     0
num_of_ratings 0
sales_price  0
discount_percent 0
sales        0
dtype: int64
```

معنی	ستون
برند موبایل	brand
مدل موبایل	model
رنگ اصلی	base_color
پردازنده	processor
سایز نمایشگر	screen_size
حافظه داخلی	ROM
حافظه RAM	RAM
اندازه واقعی صفحه (اینچ)	display_size
تعداد دوربین پشت	num_rear_camera

معنی	ستون
تعداد دوربین جلو	num_front_camera
ظرفیت باتری	battery_capacity
امتیاز کاربران	ratings
تعداد نفراتی که رای داده‌اند	num_of_ratings
قیمت فروش	sales_price
درصد تخفیف	discount_percent
Crores فروش کل، به صورت	sales

How many Mobile Brands are there

```
In [103... df["brand"].unique()
```

```
Out[103... array(['Apple', 'Poco', 'Realme', 'Samsung', 'Xiaomi'], dtype=object)
```

```
In [104... BerandDf=df.groupby('brand')
BerandDf.groups
```

```
Out[104... {'Apple': [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55], 'Poco': [56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111], 'Realme': [112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, ...], 'Samsung': [250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, ...], 'Xiaomi': [369, 370, 371, 372, 373, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405, 406, 407, 408, 409, 410, 411, 412, 413, 414, 415, 416, 417, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429]}
```

```
In [105... appledf = BerandDf.get_group("Apple")
appledf.head()
```

Out[105...

	brand	model	base_color	processor	screen_size	ROM	RAM	display_size	num_rear_c
0	Apple	iPhone SE	Black	Water	Very Small	64	2	4.7	
1	Apple	iPhone 12 Mini	Red	Ceramic	Small	64	4	5.4	
2	Apple	iPhone SE	Red	Water	Very Small	64	2	4.7	
3	Apple	iPhone XR	Others	iOS	Medium	64	3	6.1	
4	Apple	iPhone 12	Red	Ceramic	Medium	128	4	6.1	

In [106...

```
PocoDf=BerandDf.get_group('Poco')
PocoDf.head()
```

Out[106...

	brand	model	base_color	processor	screen_size	ROM	RAM	display_size	num_re
56	Poco	C3	Black	MediaTek	Large	64	4	6.5	
57	Poco	M3	Blue	Qualcomm	Large	64	4	6.5	
58	Poco	M2 Reloaded	Blue	MediaTek	Large	64	4	6.5	
59	Poco	C3	Blue	MediaTek	Large	32	3	6.5	
60	Poco	M3	Black	Qualcomm	Large	64	6	6.5	

In [107...

```
realmedf = BerandDf.get_group("Realme")
realmedf.head()
```

Out[107...

	brand	model	base_color	processor	screen_size	ROM	RAM	display_size	num_rea
112	Realme	C20	Blue	MediaTek	Large	32	2	6.5	
113	Realme	C20	Gray	MediaTek	Large	32	2	6.5	
114	Realme	C11 2021	Gray	Others	Large	32	2	6.5	
115	Realme	C11 2021	Blue	Others	Large	32	2	6.5	
116	Realme	C21Y	Black	Others	Large	64	4	6.5	


```
In [108... SamsungDf=BerandDf.get_group('Samsung')
SamsungDf.head()
```

Out[108...

	brand	model	base_color	processor	screen_size	ROM	RAM	display_size	num_r
250	Samsung	Galaxy F22	Black	MediaTek	Medium	64	4	6.4	
251	Samsung	Galaxy F22	Blue	MediaTek	Medium	64	4	6.4	
252	Samsung	Galaxy F22	Blue	MediaTek	Medium	128	6	6.4	
253	Samsung	Galaxy F22	Black	MediaTek	Medium	128	6	6.4	
254	Samsung	Galaxy F12	Blue	Exynos	Large	64	4	6.5	

```
In [109... xiaomidf = BerandDf.get_group("Xiaomi")
xiaomidf.head()
```

Out[109...

	brand	model	base_color	processor	screen_size	ROM	RAM	display_size	num_re
369	Xiaomi	Redmi 9A	Black	MediaTek	Large	32	3	6.5	
370	Xiaomi	Mi 11X	White	Qualcomm	Large	128	6	6.7	
371	Xiaomi	Redmi 8A Dual	White	Qualcomm	Medium	32	3	6.2	
372	Xiaomi	Mi A3	Blue	Qualcomm	Medium	64	4	6.1	
373	Xiaomi	Redmi 9	Blue	MediaTek	Large	128	4	6.5	

How much is count for each brand ?

```
In [110... X1=df['brand'].value_counts()
X1
```

Out[110...

```
brand
Realme      138
Samsung     119
Xiaomi       61
Apple        56
Poco         56
Name: count, dtype: int64
```

```
In [111...] appliedf['model'].value_counts()
```

```
Out[111...] model
iPhone XR          18
iPhone 12          17
iPhone 12 Mini     16
iPhone SE           3
iPhone 8            1
iPhone 7 Plus      1
Name: count, dtype: int64
```

```
In [112...] PocoDf['model'].value_counts()
```

```
Out[112...] model
M3              9
M2 Pro          9
C3              6
F3 GT           6
X3 Pro          6
X3              6
M3 Pro 5G       5
M2              4
M2 Reloaded     2
F1              2
X2              1
Name: count, dtype: int64
```

```
In [113...] realmedf['model'].value_counts()
```

```
Out[113... model
GT Master Edition    9
Narzo 30             6
8 Pro                6
7 Pro                6
X3 SuperZoom         6
3i                   6
8 5G                 6
8                    6
X7 Max               6
5 Pro                6
C11 2021             4
C2                   4
C25                  4
C21                  4
X7 5G                4
7                    4
C15                  4
Narzo 30A            4
Narzo 30 5G          4
Narzo 20             4
Narzo 30 Pro 5G      4
Narzo 10A            3
X3                   3
6                    3
GT 5G                3
C21Y                 2
C20                  2
C11                  2
C12                  2
8s 5G                2
Narzo 20 Pro         2
X7 Pro 5G            2
Narzo 20A            2
Narzo 10             2
6i                   1
Name: count, dtype: int64
```

```
In [114... SamsungDf['model'].value_counts()
```

```
Out[114... model
Galaxy A21s 7
Galaxy A03s 6
Galaxy F12 6
Galaxy F41 6
Galaxy F62 6
Galaxy A51 5
Galaxy F02s 5
Galaxy A20s 5
Galaxy A12 5
Galaxy A22 5G 4
Galaxy F22 4
Galaxy A52s 5G 4
Galaxy M02 4
Galaxy Z Flip3 5G 4
Galaxy Note 20 3
Galaxy S20 FE 3
Galaxy Z Fold3 5G 3
Galaxy A52 3
M02s 2
Galaxy J7 - 6 2
Galaxy M32 2
Galaxy A31 2
Galaxy A72 2
Galaxy A50s 2
Galaxy M01 2
Galaxy M31 2
Galaxy A71 2
Galaxy Grand 2 2
Galaxy Note 20 Ultra 5G 2
Galaxy A22 1
Galaxy M11 1
Galaxy A7 1
Galaxy J6 1
Galaxy A10 1
Galaxy Fold 2 1
Galaxy A20 1
Galaxy A80 1
Galaxy M42 1
Galaxy S21 Plus 1
Galaxy M30s 1
Galaxy Note10 Lite 1
Galaxy S21 1
Galaxy S10 1
Name: count, dtype: int64
```

Find out uniques models for each brand

```
In [115... appliedf['model'].unique()
```

```
Out[115... array(['iPhone SE', 'iPhone 12 Mini', 'iPhone XR', 'iPhone 12',
      'iPhone 8', 'iPhone 7 Plus'], dtype=object)
```

```
In [116... appliedf['model'].nunique()
```

Out[116... 6

In [117... PocoDf["model"].unique()

Out[117... array(['C3', 'M3', 'M2 Reloaded', 'X3 Pro', 'M3 Pro 5G', 'M2 Pro',
'F3 GT', 'X3', 'F1', 'M2', 'X2'], dtype=object)

In [118... PocoDf['model'].nunique()

Out[118... 11

In [119... realmedf["model"].unique()

Out[119... array(['C20', 'C11 2021', 'C21Y', 'Narzo 30 5G', 'C21', 'Narzo 30',
'8s 5G', 'Narzo 30A', '8 5G', '8 Pro', 'C15', '8',
'GT Master Edition', 'X7 5G', '7', 'Narzo 30 Pro 5G', 'C12', 'C11',
'X7 Max', 'GT 5G', '5 Pro', '3i', 'Narzo 20 Pro', '7 Pro',
'X3 SuperZoom', 'X7 Pro 5G', 'C2', 'X3', '6', '6i', 'C25',
'Narzo 20', 'Narzo 10A', 'Narzo 20A', 'Narzo 10'], dtype=object)

In [120... realmedf['model'].nunique()

Out[120... 35

In [121... SamsungDf['model'].unique()

Out[121... array(['Galaxy F22', 'Galaxy F12', 'M02s', 'Galaxy M02', 'Galaxy A22',
'Galaxy A52s 5G', 'Galaxy M32', 'Galaxy Z Flip3 5G',
'Galaxy A22 5G', 'Galaxy A21s', 'Galaxy A03s', 'Galaxy M31',
'Galaxy A51', 'Galaxy A72', 'Galaxy A12', 'Galaxy F62',
'Galaxy A31', 'Galaxy A52', 'Galaxy F02s', 'Galaxy M11',
'Galaxy F41', 'Galaxy A71', 'Galaxy Note 20', 'Galaxy Z Fold3 5G',
'Galaxy M01', 'Galaxy A50s', 'Galaxy Note 20 Ultra 5G',
'Galaxy S20 FE', 'Galaxy Grand 2', 'Galaxy Fold 2', 'Galaxy A7',
'Galaxy J6', 'Galaxy J7 - 6', 'Galaxy A10', 'Galaxy A20',
'Galaxy A80', 'Galaxy S21 Plus', 'Galaxy M42', 'Galaxy M30s',
'Galaxy A20s', 'Galaxy Note10 Lite', 'Galaxy S21', 'Galaxy S10'],
dtype=object)

In [122... SamsungDf['model'].nunique()

Out[122... 43

In [123... xiaomidf["model"].unique()

Out[123... array(['Redmi 9A', 'Mi 11X', 'Redmi 8A Dual', 'Mi A3', 'Redmi 9',
'Mi 11 Lite', 'Redmi Note 7 Pro', 'Mi 10i', 'Redmi Note 6 Pro',
'Redmi Note 9 Pro', 'Redmi Note 5 Pro', 'Redmi Note 7', 'Redmi Y3',
'Redmi K20', 'Redmi Note 4', 'Redmi Note 5', 'Mi 10T',
'Mi 10T Pro', 'Redmi Y2', 'Mi 10', 'Redmi 6 Pro', 'Mi 11X Pro 5G',
'Mi A1', 'Redmi 6A', 'Redmi 5'], dtype=object)

In [124... xiaomidf['model'].nunique()

Out[124... 25

```
In [125... df['revenue']=(df['sales']*10000000).round().astype(int)
df.head(5)
```

```
Out[125...      brand  model  base_color  processor  screen_size  ROM  RAM  display_size  num_rear_c
```

0	Apple	iPhone SE	Black	Water	Very Small	64	2	4.7	
1	Apple	iPhone 12 Mini	Red	Ceramic	Small	64	4	5.4	
2	Apple	iPhone SE	Red	Water	Very Small	64	2	4.7	
3	Apple	iPhone XR	Others	iOS	Medium	64	3	6.1	
4	Apple	iPhone 12	Red	Ceramic	Medium	128	4	6.1	



```
In [126... df['units_sold']=(df['revenue']/df['sales_price']).round().astype(int)
df.head(5)
```

```
Out[126...      brand  model  base_color  processor  screen_size  ROM  RAM  display_size  num_rear_c
```

0	Apple	iPhone SE	Black	Water	Very Small	64	2	4.7	
1	Apple	iPhone 12 Mini	Red	Ceramic	Small	64	4	5.4	
2	Apple	iPhone SE	Red	Water	Very Small	64	2	4.7	
3	Apple	iPhone XR	Others	iOS	Medium	64	3	6.1	
4	Apple	iPhone 12	Red	Ceramic	Medium	128	4	6.1	



```
In [127... df.units_sold.sum()
```

```
Out[127... np.int64(10134147)
```

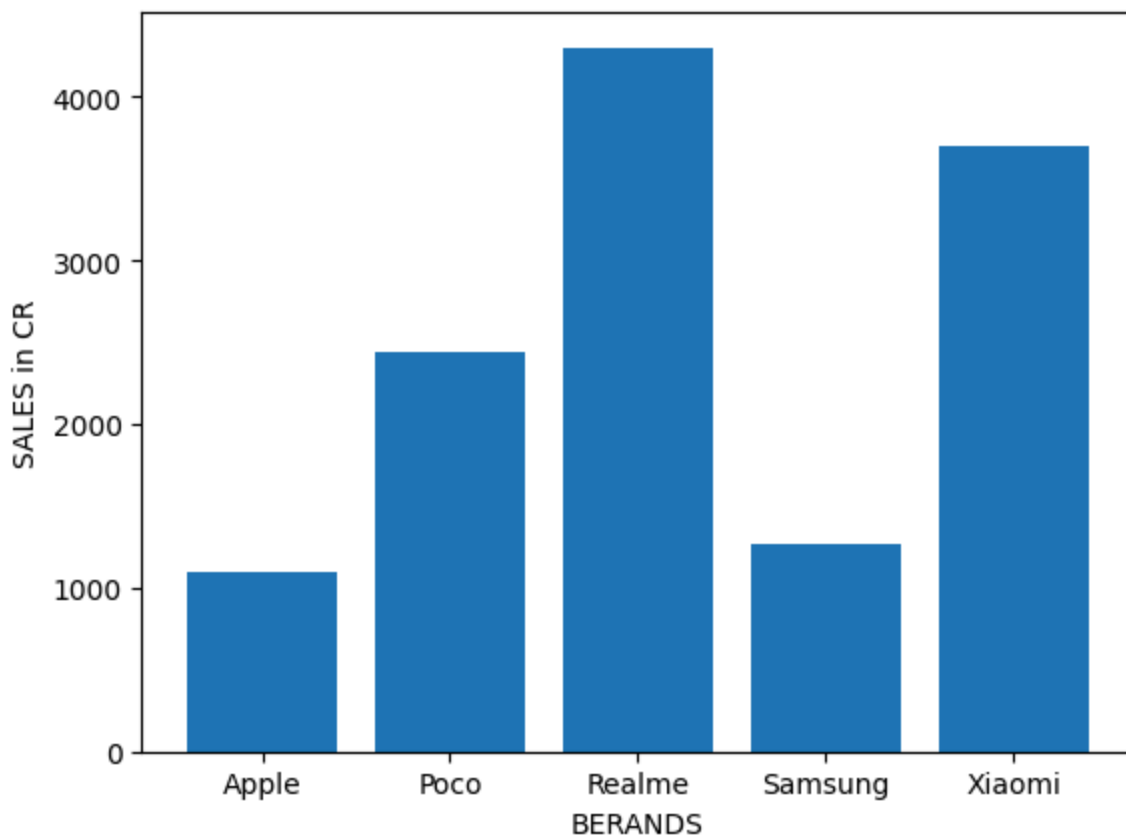
```
In [128... print('sales generated in crores',df.sales.sum(),"cr")
```

```
sales generated in crores 12793.5 cr
```

```
In [129... BerandSales=BerandDf.sales.sum()  
BerandSales
```

```
Out[129... brand  
Apple      1091.27  
Poco       2437.32  
Realme     4301.91  
Samsung    1261.90  
Xiaomi     3701.10  
Name: sales, dtype: float64
```

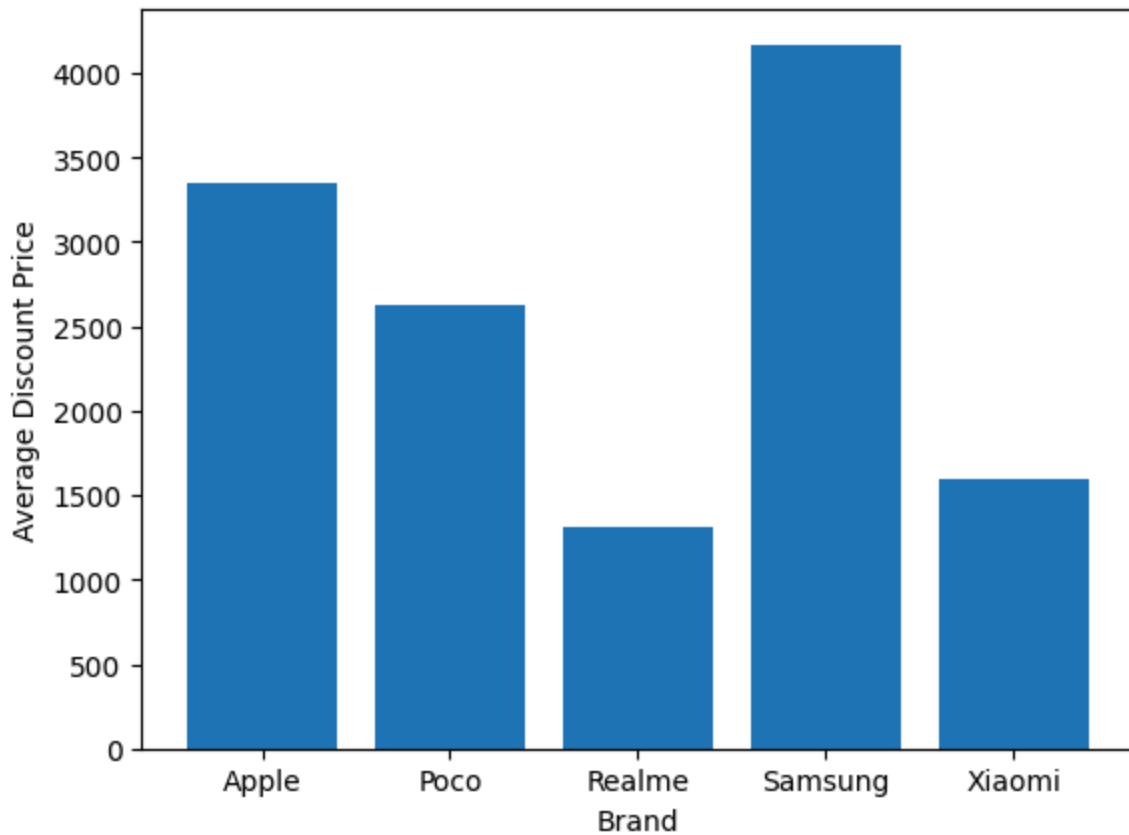
```
In [130... plt.bar(BerandSales.index,BerandSales.values)  
plt.xlabel('BERANDS')  
plt.ylabel('SALES in CR')  
plt.show()
```



```
In [131... df['actual_discount']= df['sales_price']*df['discount_percent']
```

```
In [132... avg_discount=df.groupby('brand').actual_discount.mean()
```

```
In [133... plt.bar(avg_discount.index,avg_discount.values)  
plt.xlabel('Brand')  
plt.ylabel('Average Discount Price')  
plt.show()
```

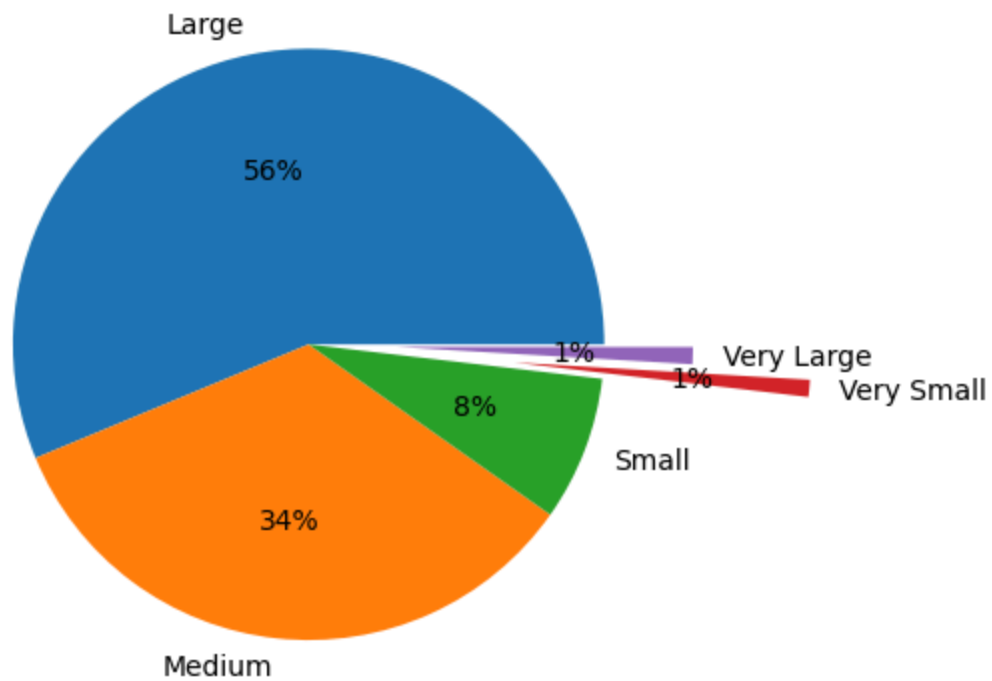


What size of display did customers like the most?

```
In [134...] screen = df.screen_size.value_counts()  
screen
```

```
Out[134...] screen_size  
Large      242  
Medium     146  
Small      34  
Very Small  4  
Very Large  4  
Name: count, dtype: int64
```

```
In [135...] plt.pie(screen, labels=screen.index, autopct='%0.1f%%', explode=[0,0,0,0.7,0.3])  
plt.show()
```

In [136...

```
for brand, brandDf in df.groupby('brand'):

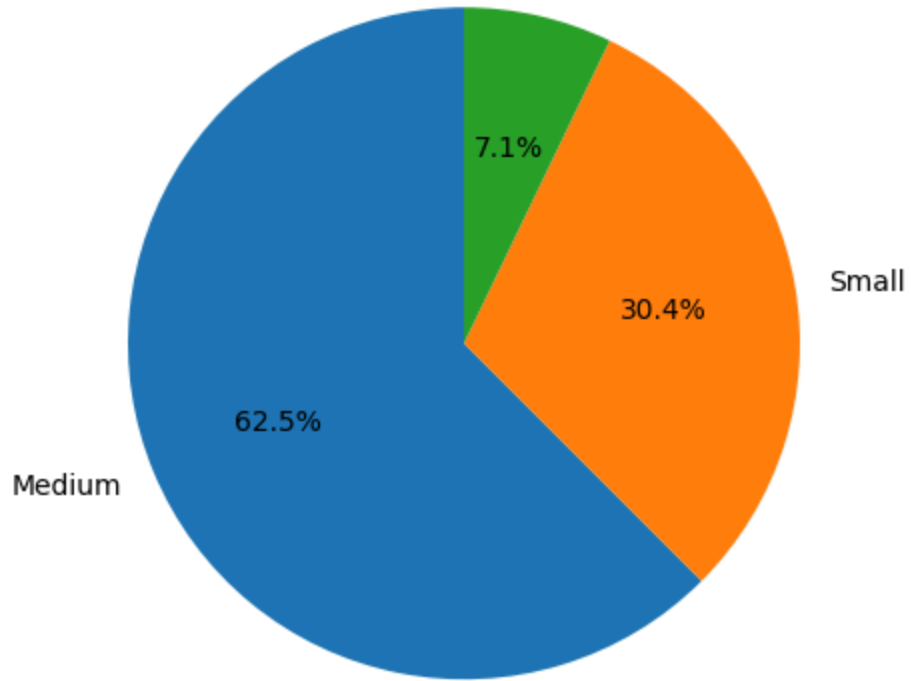
    # ستون سایز صفحه برای هر برند
    brand_screen_sizes = brandDf['screen_size']

    # شمردن سایزهای صفحه
    screen_size_counts = brand_screen_sizes.value_counts()

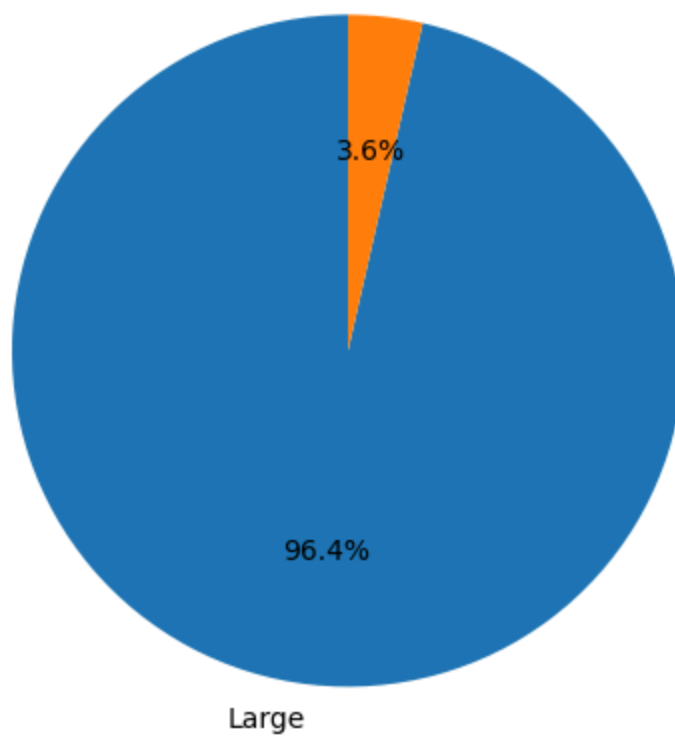
    # رسم نمودار
    plt.pie(
        screen_size_counts,
        labels=screen_size_counts.index,
        autopct='%1.1f%%',
        startangle=90
    )

    plt.title(f'Favorite Screen Size for {brand}')
    plt.axis('equal')
    plt.show()
```

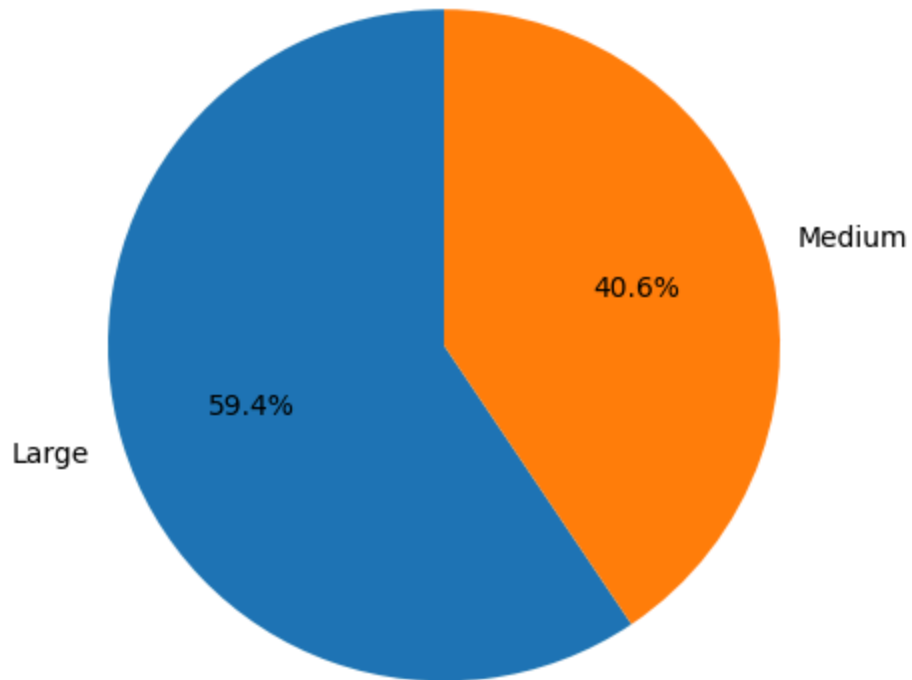
Favorite Screen Size for Apple



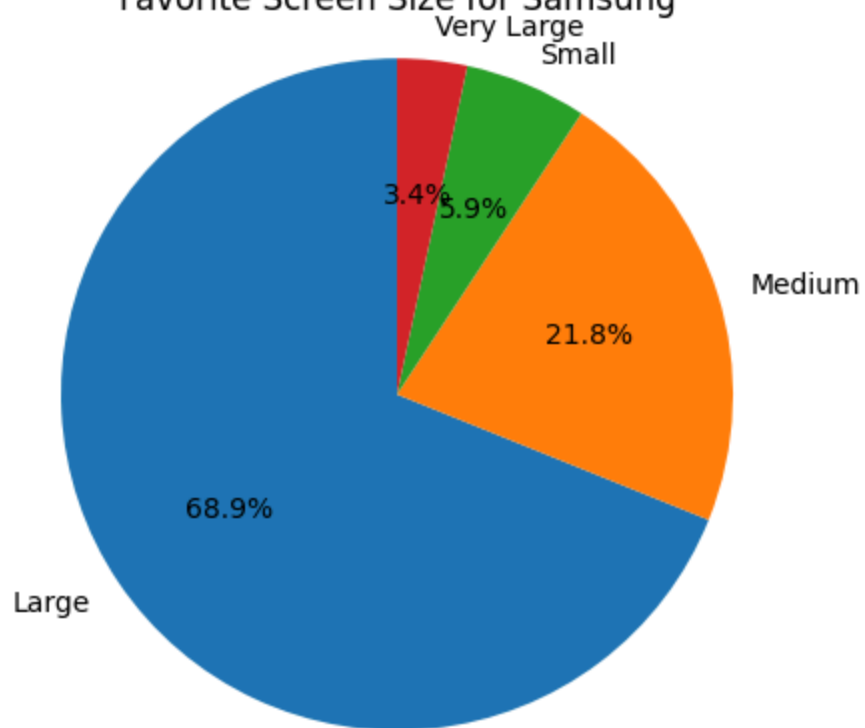
Favorite Screen Size for Poco



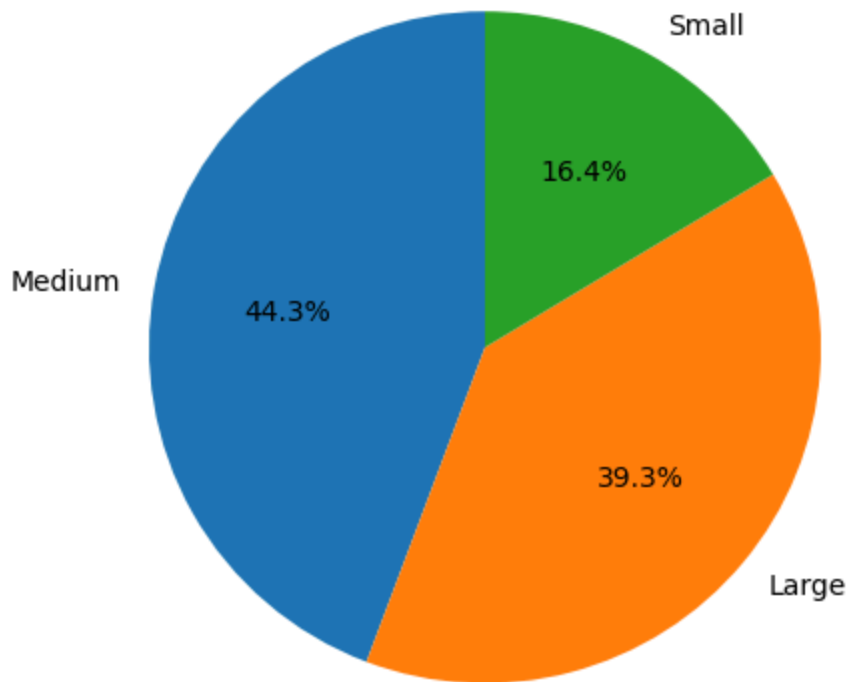
Favorite Screen Size for Realme



Favorite Screen Size for Samsung



Favorite Screen Size for Xiaomi



Which are the favorite colors of mobile phone customers?

```
In [137... Color=df['base_color'].value_counts()  
Color
```

```
Out[137... base_color  
Blue      117  
Black     112  
White      44  
Silver     32  
Others     28  
Green      24  
Red        21  
Gray       20  
Gold        11  
Yellow     11  
Purple       5  
Bronze       5  
Name: count, dtype: int64
```

```
In [138... BerandDf['base_color'].value_counts()
```

```

Out[138... brand  base_color
          Apple  Black      12
              White      11
              Blue       8
              Red        7
              Green      6
              Others     4
              Purple     4
              Yellow     3
              Gold       1
          Poco   Blue      18
              Black     17
              Yellow     5
              Gray       3
              Others     3
              Red        3
              Silver     3
              Bronze     2
              Green      2
          Realme Blue     43
              Black     26
              Silver     24
              White     16
              Gray       9
              Others     9
              Green      5
              Yellow     3
              Red        2
              Purple     1
          Samsung Black     38
              Blue     33
              Green     10
              White     10
              Others     9
              Gray       7
              Red        4
              Bronze     3
              Silver     3
              Gold       2
          Xiaomi Black     19
              Blue     15
              Gold       8
              White     7
              Red        5
              Others     3
              Silver     2
              Gray       1
              Green      1
          Name: count, dtype: int64

```

Which processors are the most favorable for customers?

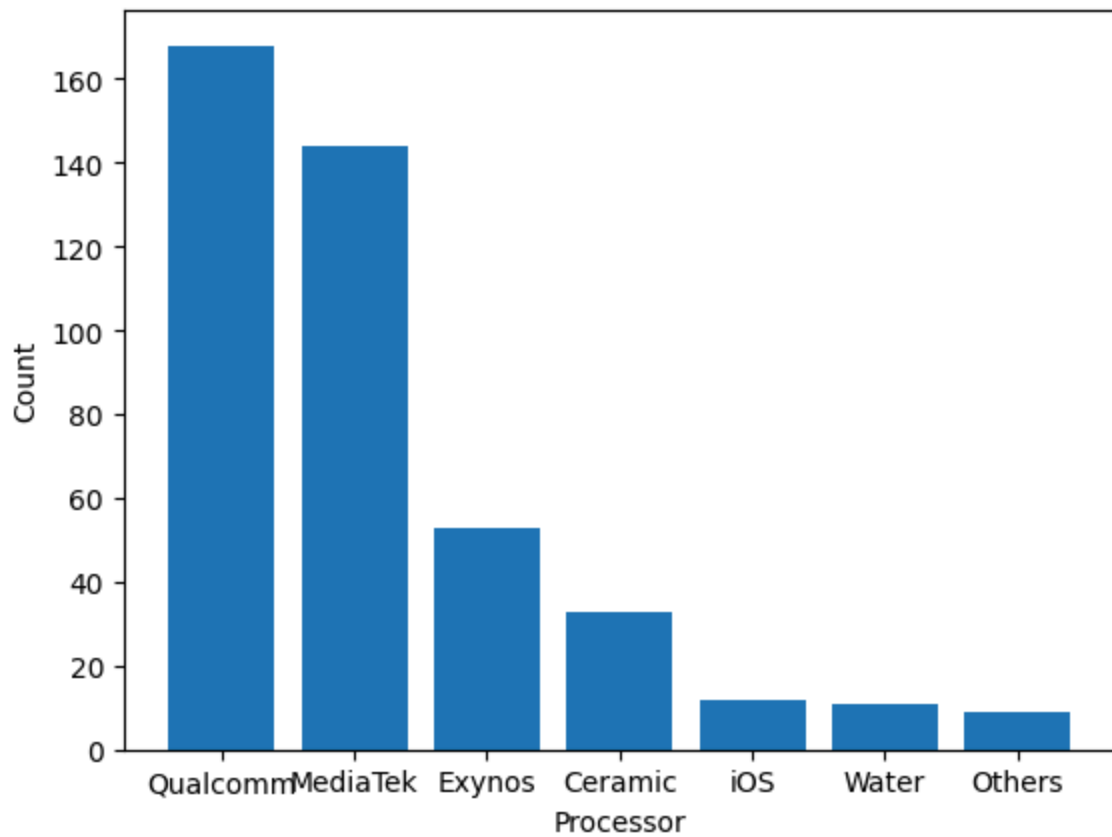
```

In [139... pro=df['processor'].value_counts()
pro

```

```
Out[139... processor
Qualcomm    168
MediaTek    144
Exynos      53
Ceramic     33
iOS         12
Water       11
Others       9
Name: count, dtype: int64
```

```
In [140... plt.bar(pro.index,pro)
plt.xlabel('Processor')
plt.ylabel('Count')
plt.show()
```



```
In [141... BerandDf.processor.value_counts()
```

```
Out[141...] brand    processor
Apple    Ceramic    33
          iOS       12
          Water     11
Poco      Qualcomm   33
          MediaTek  23
Realme    MediaTek   91
          Qualcomm  41
          Others    6
Samsung   Exynos     53
          Qualcomm  38
          MediaTek  25
          Others    3
Xiaomi    Qualcomm   56
          MediaTek   5
Name: count, dtype: int64
```

. Find out mobile phones at various price ranges. which budget range do customers choose mostly?

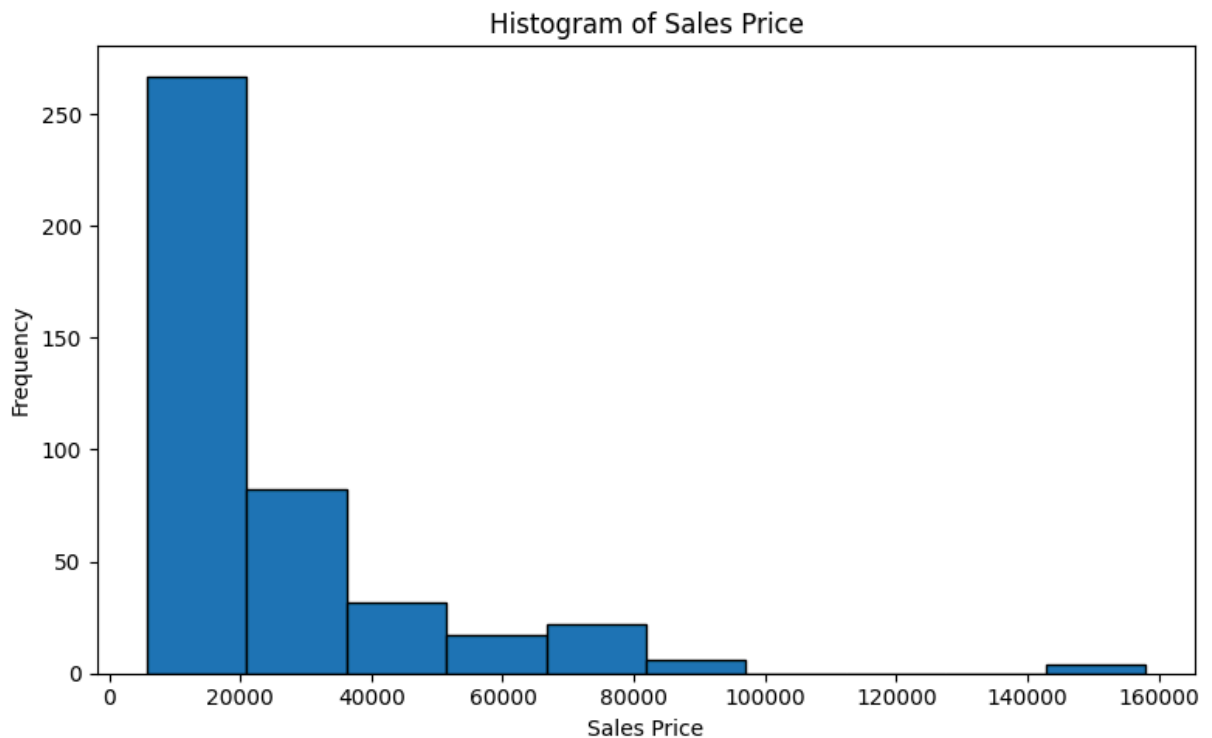
```
In [142...] df.columns
```

```
Out[142...] Index(['brand', 'model', 'base_color', 'processor', 'screen_size', 'ROM',
      'RAM', 'display_size', 'num_rear_camera', 'num_front_camera',
      'battery_capacity', 'ratings', 'num_of_ratings', 'sales_price',
      'discount_percent', 'sales', 'revenue', 'units_sold',
      'actual_discount'],
      dtype='object')
```

```
In [143...] plt.figure(figsize=(8,5))

plt.hist(
    df.sales_price,
    bins=10,          # رو مرتب می‌سازه bins خودش
    edgecolor="black" # مرزبندی واضح
)

plt.xlabel("Sales Price")
plt.ylabel("Frequency")
plt.title("Histogram of Sales Price")
plt.tight_layout()
plt.show()
```



Which are the top 10 models by avg sales

```
In [144... avg_sales_by_model=df.groupby('model')['sales'].mean()
avg_sales_by_model
```

```
Out[144... model
3i          28.195000
5 Pro       27.838333
6           72.176667
6i          78.020000
7           67.975000
...
iPhone 12 Mini    1.569375
iPhone 7 Plus    109.940000
iPhone 8         43.990000
iPhone SE       127.520000
iPhone XR        24.415000
Name: sales, Length: 119, dtype: float64
```

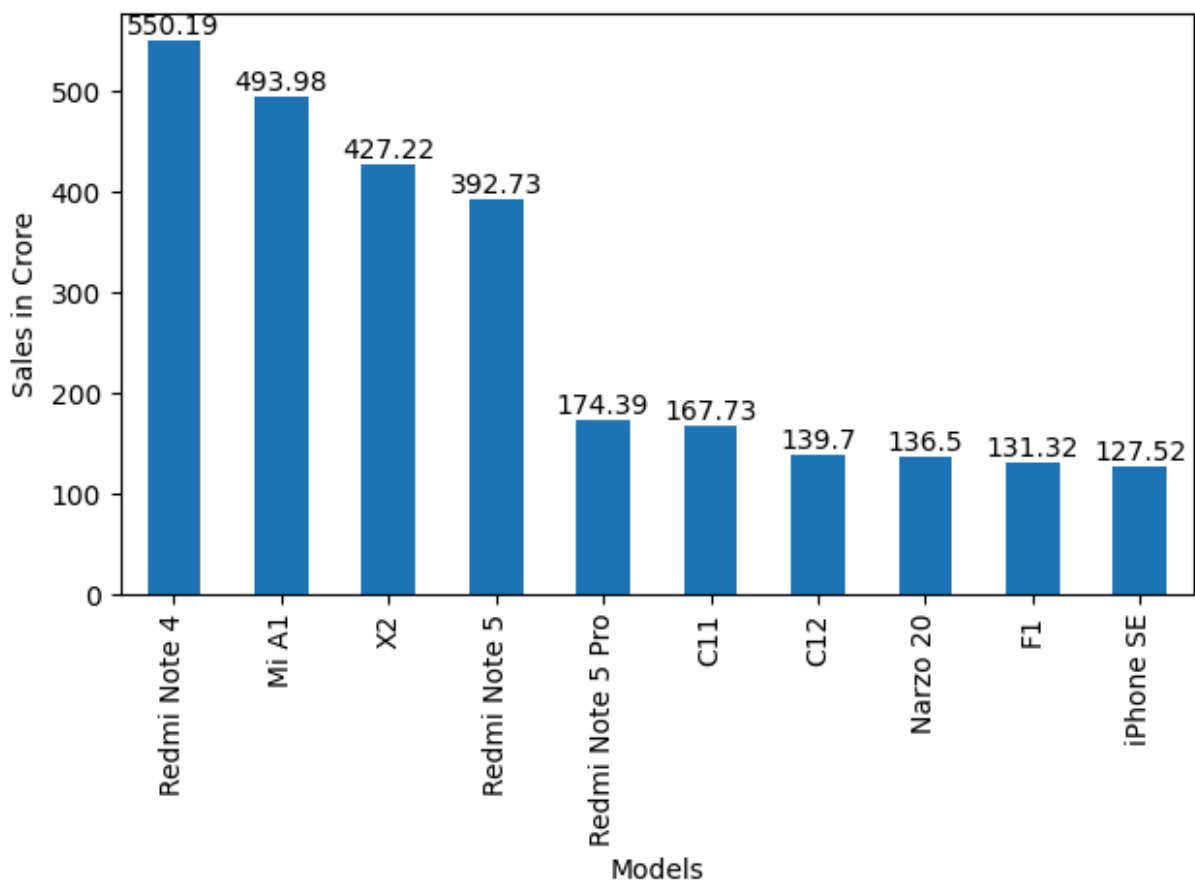
```
In [145... Top10Model=avg_sales_by_model.sort_values(ascending=False).head(10)
Top10Model
```



```
Out[145... model
Redmi Note 4      550.190
Mi A1             493.980
X2               427.220
Redmi Note 5     392.730
Redmi Note 5 Pro 174.395
C11              167.730
C12              139.700
Narzo 20         136.495
F1               131.320
iPhone SE        127.520
Name: sales, dtype: float64
```

```
In [146... Top10Model.plot(kind='bar')
plt.xlabel('Models')
plt.ylabel('Sales in Crore')
for i, value in enumerate(Top10Model):
    plt.text(i, value, str(round(value, 2)), ha='center', va='bottom')

plt.tight_layout()
plt.show()
```



What is Battery capacity by brand

```
In [147... BerandDf.battery_capacity.value_counts()
```

```

Out[147... brand    battery_capacity
          Apple    2815                33
              2942                18
              1800                 5
          Poco    5000                26
              6000                15
              5065                 6
              5160                 6
              4000                 2
              4500                 1
          Realme  5000                53
              4500                25
              6000                18
              4300                13
              4200                 9
              4035                 6
              4230                 6
              4000                 4
              4310                 4
          Samsung 5000                40
              6000                21
              4000                16
              4500                16
              3300                 7
              7000                 6
              4300                 3
              4400                 3
              2600                 2
              3400                 2
              3000                 1
              3700                 1
              4800                 1
          Xiaomi  4000                27
              5000                10
              4250                 6
              5020                 4
              4030                 3
              4520                 3
              3080                 2
              4820                 2
              3000                 1
              3300                 1
              4100                 1
              4780                 1
          Name: count, dtype: int64

```